

Technical Interview Preparation Notes: AI, Machine Learning and NLP

Section 1: Artificial Intelligence and Machine Learning Basics

What is Artificial Intelligence?

Artificial Intelligence is the field of building systems that perform tasks that normally need human intelligence, such as understanding language, recognising images, making decisions and learning from experience. Machine learning and deep learning are subsets of AI.

What is the difference between AI, Machine Learning and Deep Learning?

AI is the broad goal of making machines intelligent. Machine Learning is a subset of AI where systems learn patterns from data instead of being explicitly programmed. Deep Learning is a subset of machine learning that uses neural networks with many layers to learn complex patterns from large amounts of data.

What is Machine Learning? What are its main types?

Machine Learning lets a model learn from data to make predictions or decisions. The main types are supervised learning, unsupervised learning and reinforcement learning.

What is the difference between supervised and unsupervised learning?

Supervised learning uses labelled data, where each example has a known output, and the model learns to predict that output. Examples are classification and regression. Unsupervised learning uses unlabelled data and finds hidden structure or patterns. Examples are clustering and dimensionality reduction.

What is reinforcement learning?

In reinforcement learning an agent learns by interacting with an environment. It receives rewards for good actions and penalties for bad ones, and it learns a policy that maximises the total reward over time. It is used in game playing and robotics.

What is the difference between classification and regression?

Classification predicts a category or class, such as spam or not spam. Regression predicts a continuous numerical value, such as the price of a house.

What is overfitting in machine learning?

Overfitting happens when a model learns the training data too well, including noise, so it performs very well on training data but poorly on new unseen data. It can be reduced by using more data, simpler models, regularisation, cross-validation, dropout or early stopping.

What is underfitting?

Underfitting happens when a model is too simple to capture the pattern in the data, so it performs poorly on both training and test data. It can be fixed by using a more complex model, adding better features or training longer.

What is the bias-variance tradeoff?

Bias is the error from overly simple assumptions, which causes underfitting. Variance is the error from sensitivity to small changes in the training data, which causes overfitting. A good model balances the two. Reducing one often increases the other.

Section 2: Machine Learning Algorithms and Evaluation

How does linear regression work?

Linear regression models the relationship between input features and a continuous target using a straight line, or a hyperplane for many features. It finds the weights that minimise the mean squared error between predictions and actual values.

What is logistic regression?

Logistic regression is a classification algorithm that uses the sigmoid function to convert a linear combination of features into a probability between 0 and 1. Despite its name it is used for classification, commonly binary classification.

How does a decision tree work?

A decision tree splits the data repeatedly using feature conditions, choosing the split that best separates the classes, using measures such as Gini impurity or entropy. Each leaf gives a prediction. Decision trees are easy to interpret but can overfit.

What is a random forest?

A random forest is an ensemble of many decision trees trained on random subsets of data and features. The final prediction is the majority vote for classification or the average for regression. It reduces overfitting compared with a single tree.

What is the difference between bagging and boosting?

Bagging trains models independently in parallel on random samples and combines them, which reduces variance. Random forest is an example. Boosting trains models one after another, where each new model focuses on the mistakes of the previous one, which reduces bias. Examples are AdaBoost and XGBoost.

How does K-Means clustering work?

K-Means is an unsupervised algorithm that divides data into K clusters. It picks K initial centres, assigns each point to the nearest centre, recomputes each centre as the mean of its points, and repeats until the assignments stop changing.

What are precision, recall and F1-score?

Precision is the fraction of predicted positives that are actually positive. Recall is the fraction of actual positives that the model found. The F1-score is the harmonic mean of precision and recall and is useful when the classes are imbalanced.

What is a confusion matrix?

A confusion matrix is a table that shows true positives, true negatives, false positives and false negatives for a classifier. It is used to calculate accuracy, precision, recall and F1-score.

What is cross-validation?

Cross-validation evaluates a model by splitting the data into several folds, training on some folds and testing on the remaining fold, and repeating this so every fold is used for testing. It gives a more reliable estimate of performance than a single train-test split.

What is feature scaling and why is it needed?

Feature scaling puts features on a similar range, using normalisation or standardisation. It is important for algorithms that use distances or gradients, such as K-Nearest Neighbours, SVM and neural networks.

Section 3: Deep Learning

What is a neural network?

A neural network is a model made of layers of connected units called neurons. Each neuron takes weighted inputs, adds a bias and applies an activation function. The network has an input layer, one or more hidden layers and an output layer, and it learns by adjusting weights to reduce error.

What is an activation function? Name some common ones.

An activation function adds non-linearity so the network can learn complex patterns. Common examples are ReLU, sigmoid, tanh and softmax. ReLU is widely used in hidden layers, and softmax is used in the output layer for multi-class classification.

What is backpropagation?

Backpropagation is the algorithm used to train neural networks. It calculates the error at the output and propagates it backward through the layers using the chain rule to compute the gradient of the loss with respect to each weight. The weights are then updated using gradient descent.

What is gradient descent?

Gradient descent is an optimisation algorithm that updates model parameters in the direction that reduces the loss function. The learning rate controls the size of each step. Variants include batch, stochastic and mini-batch gradient descent, and optimisers such as Adam.

What is a CNN and where is it used?

A Convolutional Neural Network uses convolution filters to detect local patterns such as edges and shapes in images. It is mainly used for image classification, object detection and other computer vision tasks.

What is an RNN and what is its limitation?

A Recurrent Neural Network processes sequences by passing a hidden state from one step to the next, so it is used for text and time series. Its limitation is the vanishing gradient problem, which makes it hard to learn long-range dependencies. LSTM and GRU were designed to reduce this problem.

What is dropout?

Dropout is a regularisation technique where a random fraction of neurons is ignored during each training step. It prevents the network from depending too much on particular neurons and reduces overfitting.

Section 4: Natural Language Processing

What is NLP?

Natural Language Processing is the field of AI that enables computers to understand, interpret and generate human language. Applications include chatbots, translation, sentiment analysis, summarisation and question answering.

What is tokenization?

Tokenization splits text into smaller units called tokens, such as words, subwords or characters. It is the first step in most NLP pipelines. Modern language models often use subword tokenization.

What are stop words, stemming and lemmatization?

Stop words are very common words such as the, is and and, which are often removed. Stemming cuts words down to a root form by removing endings, and it can produce non-words, for example studies becomes studi. Lemmatization converts a word to its proper dictionary form using vocabulary and grammar, for example better becomes good. Lemmatization is more accurate but slower.

What is TF-IDF?

TF-IDF stands for Term Frequency - Inverse Document Frequency. It scores how important a word is to a document within a collection. Term frequency counts how often the word appears in the document, and inverse document frequency reduces the weight of words that appear in many documents.

What are word embeddings?

Word embeddings represent words as dense numerical vectors so that words with similar meanings have similar vectors. Examples are Word2Vec, GloVe and FastText. They capture semantic relationships, unlike one-hot encoding or bag-of-words.

What is the difference between bag-of-words and embeddings?

Bag-of-words counts word occurrences and ignores order and meaning, producing sparse high-dimensional vectors. Embeddings produce dense, low-dimensional vectors that capture semantic similarity.

What is the Transformer architecture?

The Transformer is a neural network architecture based on self-attention, introduced in the paper Attention Is All You Need. Self-attention lets each token look at all other tokens in the sequence, so the model captures long-range context and can be trained in parallel. Transformers are the basis for BERT, GPT and most modern language models.

What is the attention mechanism?

Attention lets a model focus on the most relevant parts of the input when producing each output. It computes weights that show how much each token should contribute, using query, key and value vectors.

What is the difference between BERT and GPT?

BERT is an encoder-only model that reads text in both directions and is mainly used to understand text, for tasks such as classification and question answering. GPT is a decoder-only model that predicts the next token from left to right and is mainly used to generate text.

What are common NLP evaluation tasks and metrics?

Classification tasks use accuracy, precision, recall and F1-score. Machine translation often uses BLEU, summarisation uses ROUGE, and language models are measured with perplexity.

Section 5: Generative AI, LLMs and RAG

What is a Large Language Model?

A Large Language Model is a Transformer-based model trained on very large amounts of text to predict the next token. It can answer questions, summarise, translate and write code. Examples include GPT, Gemini, Claude and Llama.

What is prompt engineering?

Prompt engineering is the practice of designing clear instructions and context for a language model to get better outputs. Techniques include giving a role, adding examples (few-shot prompting), asking for step-by-step reasoning and specifying the output format.

What is hallucination in LLMs?

Hallucination is when a language model produces fluent but incorrect or made-up information. It can be reduced by grounding the model in retrieved documents, as in RAG, by instructing it to answer only from the context, and by verifying outputs.

What is RAG (Retrieval-Augmented Generation)?

RAG combines retrieval and generation. The system first retrieves relevant information from an external knowledge source and then gives that information to the language model as context so it can generate an answer based on it. This reduces hallucination and lets the model use private or up-to-date data without retraining.

What are the steps of a RAG pipeline?

Documents are loaded with a document loader and split into chunks. Each chunk is converted into an embedding vector and stored in a vector database. When a user asks a question, the question is embedded, the most similar chunks are retrieved, and they are passed with the question to the language model, which generates the answer.

What is a vector database? Name some examples.

A vector database stores embedding vectors and supports fast similarity search to find the vectors closest to a query vector. Examples are FAISS, ChromaDB, Pinecone and Weaviate.

What is the difference between FAISS and ChromaDB?

FAISS is a library from Meta for fast similarity search over vectors and focuses on the index itself. ChromaDB is a vector database that stores vectors together with documents and metadata, persists data easily and supports metadata filtering.

What is chunking and why is it needed in RAG?

Chunking splits large documents into smaller pieces so retrieval can find the specific part that answers the question. It keeps the context within the language model limit and improves retrieval precision. Chunk overlap preserves context across chunk boundaries.

What is fine-tuning and how is it different from RAG?

Fine-tuning further trains a model on specific data, which changes the model weights and teaches it a style or skill. RAG does not change the model. It supplies relevant information at query time. RAG is better for frequently changing knowledge and for citing sources.

What are advanced RAG techniques?

Common improvements include hybrid retrieval, which combines keyword search such as BM25 with vector search, query rewriting or multi-query generation, reranking the retrieved chunks with a more accurate model, metadata filtering and contextual compression.